higher education & training

Department:
Higher Education and Training
REPUBLIC OF SOUTH AFRICA

NATIONAL CERTIFICATE (VOCATIONAL)

CONSTRUCTION PLANNING
NQF LEVEL 3

(12010253)

12 November 2019 (X-Paper)
09:00–13:00

This question paper consists of 5 pages and 1 diagram sheet.

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TIME: 4 HOURS
MARKS: 100

INSTRUCTIONS AND INFORMATION

1. Answer ALL the questions.
2. Read ALL the questions carefully.
3. Number the answers according to the numbering system used in this question paper.
4. The drawings, sketches and diagrams must be done in pencil.
5. The sketches and/or diagrams must be neat, reasonably large, in proportion, and fully labelled.
6. Rule off across the page after each completed question.
7. ALL the abbreviations and symbols must comply with the latest national Building Regulations and all relevant SANS (SABS) codes.
8. QUESTION 2.2 and 2.3 must be answered on the DRAWING PAPER.
9. ALL other answers must be done in the ANSWER BOOK.
10. Start each section on a NEW page.
11. Use only BLUE or BLACK ink.
12. Write neatly and legibly.

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QUESTION 1

1.1 Various options are given as possible answers to the following questions. Choose the correct answer and write only the letter (A–C) next to the question number (1.1.1–1.1.5) in the ANSWER BOOK.

1.1.1 The main function of the architect is to …

A regulate buildings.
B interpret the ideas of the client.
C give instructions.

1.1.2 Who must communicate with the contractor through the drawings and plans to ensure that the contractor knows what to build?

A Project manager.
B Design team.
C Architect.

1.1.3 What is the aim of clear communication?

A To work safely and achieve excellent standards in the shortest time.
B That is how people work together on a project.
C To interpret the drawings.

1.1.4 Regarding the approval of building plans, what other party is involved in the communication process?

A South African Bureau of Standards.
B National building regulations.
C Local authority or municipality.

1.1.5 Regarding communication channels, who chairs meetings on site on behalf of the client?

A Architect.
B Contractor.
C Site foreman.

(5 × 1) (5)

1.2 Give THREE examples of physical barriers to communication on a building site. (3)

1.3 State ONE advantage of each of the following examples of non-verbal communication:

1.3.1 Showing by way of examples

1.3.2 Simplified instruction

(2 × 1) (2)

[10]

QUESTION 2

2.1 List THREE important facts about construction drawings. (3)

2.2 A 30°-pitch galvanised corrugated-iron roof on 75 mm × 50 mm purlins is constructed of 114 mm × 38 mm rafters, tie beam and wall plates with an overhang of 300 mm. It is supported by a 270 mm thick face cavity wall which is reduced to 220 mm by means of a plinth brick only four courses below the wall plate.

The wall has a 19 mm plaster on the inside. A steel door frame forms part of the cavity wall. The overhang has closed eaves with 50 mm × 25 mm slats secured to 76 mm × 50 mm timber brackets. The sprocket ends are concealed by a 228 mm × 32 mm fascia board with a 100 mm × 100 mm galvanised gutter and downpipe in position.

The ceiling is constructed of 1 000 mm × 400 mm × 15 mm thick acoustic panelling secured to the 38 mm × 38 mm brandering by means of a 3 mm thick aluminium H-section with a 75 mm × 19 mm cornice.

Draw to a scale of 1:10 a vertical section through the eaves and cavity wall. Show approximately 500 mm of the top of the double rebated steel door frame according to the national standards. (15)

2.3 FIGURE 1, DIAGRAM SHEET 1 (attached) shows an isometric view of a block. Do NOT copy the views but draw to a scale of 1:1, in first-angle orthographic projection, the following views:

- A front view in the direction of arrow A
- A top view
- A left view

NOTE: Do NOT add any dimensions.
Add hidden detail to all views.
Give the title to each view. (12)

[30]

QUESTION 3

3.1 The paragraph below explains the responsibilities of an architect and quantity surveyor as stakeholders in the construction industry.

Complete the following paragraph by using the words given in the list below. Write only the word(s) next to the question number (3.1.1–3.1.7) in the ANSWER BOOK.

regulations; tender documents; local authority; site inspections; bill of quantities; design; working drawings

The client employs the architect to (3.1.1 …) the structure. The architect must ensure that the structure complies with the (3.1.2 …) set by the (3.1.3 …). Then the architect has to develop the (3.1.4 …) which the client has agreed to. The quantity surveyor then determines the quantities of materials by using the (3.1.5 …) to prepare the (3.1.6 …) and control the building costs. After the tender has been awarded, the architect does the (3.1.7 …) to ensure that the quality of workmanship complies with the stipulations in the tender document. (7)

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3.2 A number of organisations regulate the South African construction industry.

Give the full names of SIX regulatory bodies in South Africa. (6 × 2) (12)

3.3 Explain the functions of the following trade organisations in the building and civil industries:

3.3.1 Hot Deep Galvanisers' Association of South Africa

3.3.2 Master Builders' Association

3.3.3 Clay Brick Association of South Africa

(3 × 3) (9)

3.4 Define the term *trade organisation*. (2)
[30]

QUESTION 4

4.1 Building regulations and legislation are necessary because they set guidelines for the construction industry in South Africa.

Explain the rationale behind building regulations and legislation in the construction industry. (10)

4.2 Name and explain FIVE building regulations in the 282 laws contained in the Code of Hammurabi. (5 × 2) (10)

4.3 List different types of tests that are performed on a product before Agreement South Africa issues a certificate that declares it fit for its purpose. (8)

4.4 State the purpose of the NHBRC. (2)
[30]

TOTAL: 100

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DIAGRAAM SHEET

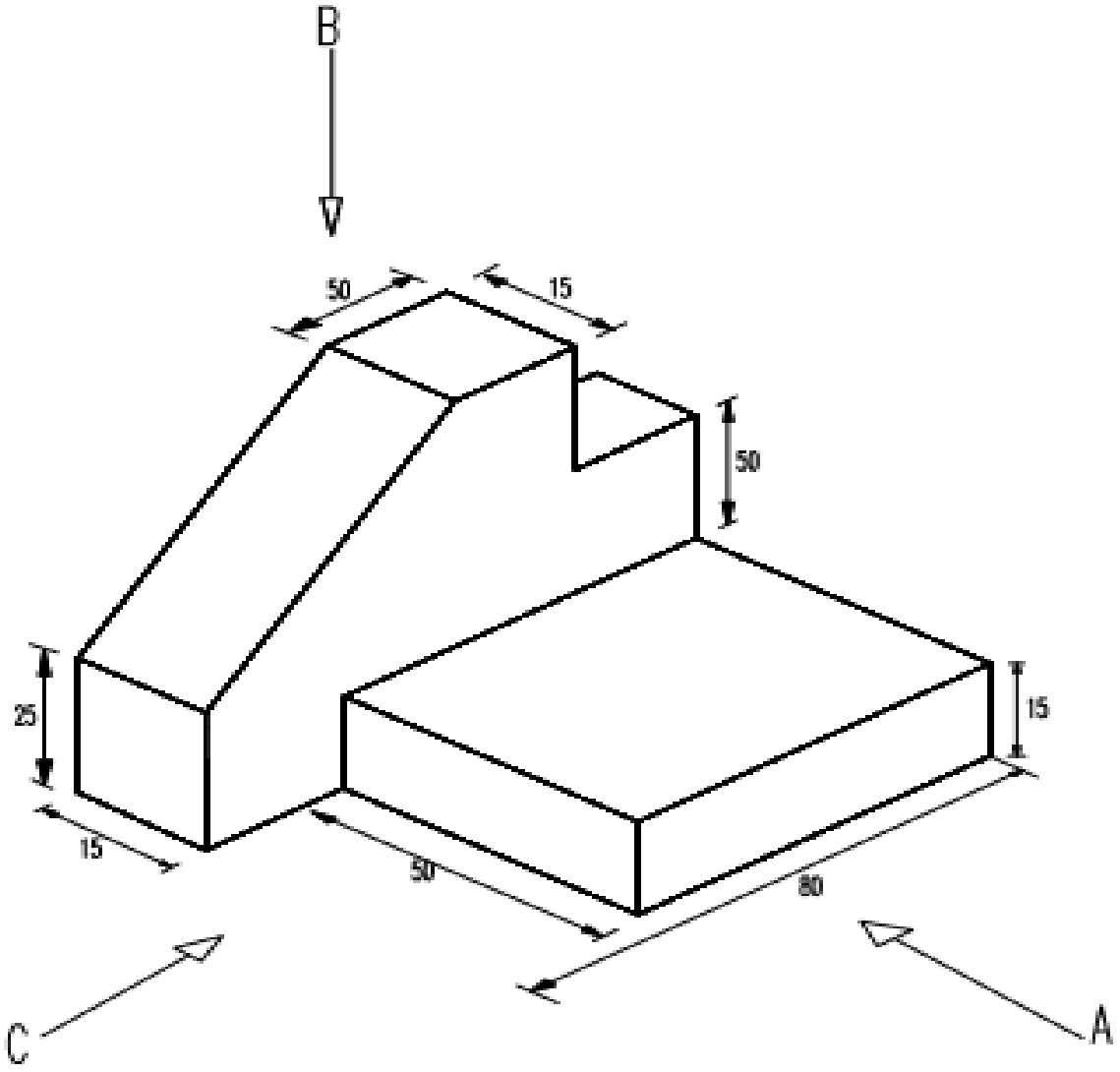


FIGURE 1